Activity 9 - Evaluate Uncertainty and Product Benchmark Performance

Alignment: K6 | K7 | A5 | A6 | LO4 Duration: 45 minutes Objective: Determine material uncertainty factors and compare the Lumina 400 with credible, method-aligned benchmarks.

Scenario

The corrected Lumina 400 result is 186 kg CO2e per functional unit. Supplier aluminium data varies by ±25%, China grid factors differ by region, customer operating hours range widely and end-of-life recycling evidence is weak. Marketing compares the result with three competitor declarations: 175, 210 and 245 kg CO2e, but their functional units, lifetimes, boundaries and PCR editions differ.

Materials

- Activity 8 model and contribution results
- Input ranges and data-quality ratings
- Three fictional competitor EPD extracts
- Uncertainty and benchmark workbook

Questions

- Which uncertainty factors could change the decision?
- What sensitivity or scenario tests are proportionate?
- Are the competitor results comparable?
- What claim, if any, can HarbourLight support?

Detailed procedure

1. Identify activity-data, factor, model, sampling, scenario and boundary-judgement uncertainty.
2. Rate likelihood, range, direction, evidence quality and decision impact for each factor.
3. Select high-impact weak inputs for sensitivity testing and document plausible ranges and sources.
4. Run one-at-a-time sensitivities and coherent scenarios for aluminium, grid, lifetime and recycling.
5. Determine whether ranges change hotspot ranking, total result, target status or proposed claim.
6. Screen each benchmark for functional unit, product performance, system boundary, allocation, geography, technology, GWP basis and vintage.
7. Exclude non-comparable records or normalise only where a defensible method and evidence exist.
8. Present ranges, comparability limitations and ranking without false precision.
9. Recommend approve, qualify, defer or reject the proposed product claim and state evidence needed for reconsideration.

Required evidence

- Uncertainty register
- Sensitivity and scenario results

- Benchmark comparability screen
- Performance and claim recommendation

Acceptance check

All material uncertainty classes are addressed; sensitivity links to decisions; only genuinely comparable benchmarks influence the conclusion; claim wording matches evidence and limitations.

Files

- Activity-09-Workbook.xlsx - editable working file
- Activity-09-Activity-Pack.pdf - printable scenario, questions, procedure and checklist

Use fictional, anonymised or authorised workplace information only.